

43.2 B-splines

Chapter 42 fixed the space we shall work in. Given a degree $d \geq 0$, an interval $[a, b]$ and K interior knots $a < t_1 < \dots < t_K < b$, the **spline space** is the set of functions that are polynomials of degree at most d on each interval between consecutive knots and have $d - 1$ continuous derivatives at each knot (Definition 42.3.1). Its dimension is $d + 1 + K$, and the truncated power functions $1, x, \dots, x^d, (x - t_1)_+^d, \dots, (x - t_K)_+^d$ are a basis (Theorem 42.3.1). (Chapter 42 writes κ_i for these knots; here κ is needed for the longer sequence built next.)

That basis is a bad one to compute with, for the reason Proposition 42.4.1 gives: its members are global, nearly collinear, and of wildly different sizes once the knots are close together. This section builds the basis everything else in the chapter uses. It spans the same space, so nothing about the *fit* changes; only the numbers do, and they change dramatically.

Warning. From here to the end of the chapter K is the number of interior knots, not a kernel; the kernel $K(\cdot)$ of Section 43.1 does not reappear.

43.2.1 The knot sequence and the recurrence

B-splines are defined by a recurrence on the degree, and the recurrence needs knots outside $[a, b]$ as well as inside it. Collect the interior knots and $d + 1$ copies of each endpoint into a nondecreasing **extended knot sequence** $\kappa_1 \leq \kappa_2 \leq \dots \leq \kappa_{m+d+1}$ with $m = K + d + 1$, namely

$$\begin{aligned} \kappa_1 = \dots = \kappa_{d+1} &= a, & \kappa_{d+1+i} &= t_i \quad (i = 1, \dots, K), \\ \kappa_{d+K+2} = \dots = \kappa_{m+d+1} &= b. \end{aligned} \tag{43.2.1}$$

This is the **clamped** sequence; an alternative is discussed below. In either case there are $m + d + 1 = K + 2d + 2$ entries and, as we shall see, $m = K + d + 1$ B-splines, exactly the dimension of the spline space. From here on κ_j always means an entry of the extended sequence.

Definition 43.2.1 — B-splines

Given a nondecreasing sequence $\kappa_1 \leq \dots \leq \kappa_{m+d+1}$, define

$$B_j^0(x) = \begin{cases} 1, & \kappa_j \leq x < \kappa_{j+1}, \\ 0, & \text{otherwise,} \end{cases} \quad j = 1, \dots, m + d,$$

and recursively, for $r = 1, \dots, d$ and $j = 1, \dots, m + d + 1 - r$,

$$B_j^r(x) = \omega_j^r(x) B_j^{r-1}(x) + \{1 - \omega_{j+1}^r(x)\} B_{j+1}^{r-1}(x), \tag{43.2.2}$$

where $\omega_j^r(x) = (x - \kappa_j) / (\kappa_{j+r} - \kappa_j)$.

A term whose denominator vanishes is read as zero. The functions $B_1^d, \dots, B_m^d$ are the **B-splines of degree d** on the sequence, and the matrix $\mathbf{B}$ with $\mathbf{B}_{ij} = B_j^d(x_i)$ is the **B-spline basis matrix** of a design.

The recurrence is due to de Boor (1972) and Cox (1972). It has an immediate reading: B_j^r is a weighted average of two B-splines of one degree lower, with weights that ramp linearly from 0 to 1 across the support. The piece of the definition that does the work is the convention about zero denominators, which is what lets the endpoints be repeated; a repeated knot makes the corresponding lower-degree B-spline identically zero, and the offending term disappears with it.

43.2.2 What B-splines are like

Proposition 43.2.1 — Properties of the B-spline basis

Let $B_1^d, \dots, B_m^d$ be as in Definition 43.2.1. Then:

- (a) (*Local support.*) $B_j^d(x) = 0$ for $x \notin [\kappa_j, \kappa_{j+d+1})$, and, for the sequence Eq. 43.2.1 with distinct interior knots, $B_j^d(x) > 0$ for $\kappa_j < x < \kappa_{j+d+1}$. In particular at most $d + 1$ of the B_j^d are nonzero at any point.
- (b) (*Nonnegativity.*) $B_j^d \geq 0$ everywhere.
- (c) (*Partition of unity.*) For the sequence Eq. 43.2.1, $\sum_{j=1}^m B_j^d(x) = 1$ for every $x \in [a, b)$.
- (d) (*Piecewise polynomial.*) On each interval between consecutive distinct knots, B_j^d is a polynomial of degree at most d .
- (e) (*Derivative.*) For $d \geq 1$,

$$\frac{d}{dx} B_j^d(x) = d \left\{ \frac{B_j^{d-1}(x)}{\kappa_{j+d} - \kappa_j} - \frac{B_{j+1}^{d-1}(x)}{\kappa_{j+d+1} - \kappa_{j+1}} \right\}, \quad (43.2.3)$$

with the same convention on zero denominators. Consequently B_j^d has $d - 1$ continuous derivatives at every simple knot.

- (f) (*Basis.*) For the sequence Eq. 43.2.1 with distinct interior knots, $B_1^d, \dots, B_m^d$ are linearly independent on $[a, b]$ and span the spline space of Theorem 42.3.1, of which they are therefore a basis. (With a repeated interior knot they span a larger space, one with a weaker continuity requirement there; see Exercise (C1).)

Proof.

- (a) and (b) go together, by induction on r . For $r = 0$ both are clear. Suppose they hold for $r - 1$. By Eq. 43.2.2, B_j^r is a combination of B_j^{r-1} , supported in $[\kappa_j, \kappa_{j+r})$, and B_{j+1}^{r-1} , supported in $[\kappa_{j+1}, \kappa_{j+r+1})$, so B_j^r vanishes outside $[\kappa_j, \kappa_{j+r+1})$. On that interval $\omega_j^r(x) = (x - \kappa_j)/(\kappa_{j+r} - \kappa_j) \geq 0$ and $1 - \omega_{j+1}^r(x) = (\kappa_{j+r+1} - x)/(\kappa_{j+r+1} - \kappa_{j+1}) \geq 0$, so $B_j^r \geq 0$. For strict positivity, let $\kappa_j < x < \kappa_{j+r+1}$. If $x < \kappa_{j+r}$ then $\kappa_j < \kappa_{j+r}$, so $\omega_j^r(x) > 0$ and $B_j^{r-1}(x) > 0$ by the induction hypothesis, and the second term is nonnegative. If $x \geq \kappa_{j+r}$ then $\kappa_{j+1} < \kappa_{j+r+1}$ — otherwise $\kappa_{j+r} = \kappa_{j+r+1}$, which contradicts $\kappa_{j+r} \leq x < \kappa_{j+r+1}$ — so $1 - \omega_{j+1}^r(x) > 0$ and $B_{j+1}^{r-1}(x) > 0$. Finally, a point lies in at most $d + 1$ of the intervals $[\kappa_j, \kappa_{j+d+1})$.
- (b) Induct on r , proving $\sum_{j=1}^{m+d+1-r} B_j^r(x) = 1$ for $x \in [a, b)$. For $r = 0$ the intervals $[\kappa_j, \kappa_{j+1})$ that are nonempty partition $[a, b)$. For the step, substitute Eq. 43.2.2

and collect terms with the same lower index:

$$\begin{aligned} \sum_j B_j^r &= \omega_1^r B_1^{r-1} + \sum_{j \geq 2} \{\omega_j^r + 1 - \omega_j^r\} B_j^{r-1} \\ &\quad + \{1 - \omega_{L+1}^r\} B_{L+1}^{r-1}, \end{aligned}$$

where $L = m + d + 1 - r$ is the largest index of a degree- r B-spline. The first and last terms vanish on $[a, b)$: for Eq. 43.2.1, B_1^{r-1} is supported in $[\kappa_1, \kappa_r) = [a, a)$, which is empty when $r \leq d$, and B_{L+1}^{r-1} in $[\kappa_{L+1}, \kappa_{L+r}) = [b, b)$. What is left is $\sum_{j=2}^L B_j^{r-1}$, which is the partition of unity one degree lower.

- (c) By induction: B_j^0 is constant on each such interval, and Eq. 43.2.2 multiplies polynomials of degree $r - 1$ by linear functions.
- (d) Induct on d . Write $D_j^r = (\kappa_{j+r} - \kappa_j)^{-1}$, so that $\omega_j^r(x) = (x - \kappa_j)D_j^r$ and $\{\omega_j^r\}' = D_j^r$. For $d = 1$, B_j^1 is continuous and piecewise linear, and differentiating Eq. 43.2.2 on the interior of each knot interval gives $D_j^1 B_j^0 - D_{j+1}^1 B_{j+1}^0$, which is Eq. 43.2.3. For the step, differentiate Eq. 43.2.2:

$$\begin{aligned} (B_j^d)' &= D_j^d B_j^{d-1} - D_{j+1}^d B_{j+1}^{d-1} \\ &\quad + \omega_j^d (B_{j+1}^{d-1})' + \{1 - \omega_{j+1}^d\} (B_{j+1}^{d-1})', \end{aligned}$$

so it suffices to show that the last two terms equal $(d - 1)\{D_j^d B_j^{d-1} - D_{j+1}^d B_{j+1}^{d-1}\}$. Substituting the induction hypothesis on the left and Eq. 43.2.2 on the right, both sides become combinations of B_j^{d-2} , B_{j+1}^{d-2} and B_{j+2}^{d-2} , and it is enough to match the three coefficients. For B_j^{d-2} : $\omega_j^d D_j^{d-1} = (x - \kappa_j)D_j^d D_j^{d-1}$ on the left and $D_j^d \omega_j^{d-1} = D_j^d (x - \kappa_j)D_j^{d-1}$ on the right. For B_{j+1}^{d-2} : $-\{1 - \omega_{j+1}^d\} D_{j+2}^{d-1} = -(\kappa_{j+d+1} - x)D_{j+1}^d D_{j+2}^{d-1}$ on the left, and $-D_{j+1}^d \{1 - \omega_{j+2}^{d-1}\} = -D_{j+1}^d (\kappa_{j+d+1} - x)D_{j+2}^{d-1}$ on the right. For B_{j+1}^{d-2} the two coefficients are $D_{j+1}^{d-1} \{1 - \omega_j^d - \omega_{j+1}^d\}$ and $D_j^d - \omega_{j+1}^{d-1} (D_j^d + D_{j+1}^d)$, and their difference reduces, after cancelling $(x - \kappa_{j+1})D_{j+1}^{d-1} D_{j+1}^d$ and collecting the terms in D_j^d , to $D_j^d \{D_{j+1}^{d-1} (\kappa_{j+d} - \kappa_{j+1}) - 1\}$, which is zero because $D_{j+1}^{d-1} = (\kappa_{j+d} - \kappa_{j+1})^{-1}$. Smoothness now follows by a second induction: B_j^1 is continuous, and if every B-spline of degree $d - 1$ is C^{d-2} at simple knots, then Eq. 43.2.2 makes B_j^d at least C^{d-2} , while Eq. 43.2.3 makes its derivative a combination of C^{d-2} functions, so $B_j^d \in C^{d-1}$.

- (e) By (d) and (e) each B_j^d belongs to the spline space, which by Theorem 42.3.1 has dimension $d + 1 + K = m$. Since there are m of them, it is enough to prove independence, and we do so by induction on d . For $d = 0$, $m = K + 1$ and the B_j^0 are indicators of disjoint intervals. Let $d \geq 1$ and suppose $s = \sum_{j=1}^m c_j B_j^d \equiv 0$ on $[a, b]$. By Eq. 43.2.3, with $c_0 = c_{m+1} = 0$,

$$s' = d \sum_{j=2}^m \frac{c_j - c_{j-1}}{\kappa_{j+d} - \kappa_j} B_j^{d-1}, \quad (43.2.4)$$

the terms $j = 1$ and $j = m + 1$ having vanished because $\kappa_{1+d} = \kappa_1 = a$ and $\kappa_{m+1+d} = \kappa_{m+1} = b$ for the sequence Eq. 43.2.1. The knots $\kappa_2, \dots, \kappa_{m+d}$ are the clamped sequence of degree $d - 1$ for the same K interior knots, and $B_2^{d-1}, \dots, B_m^{d-1}$ are its $K + d = m - 1$ B-splines. By the induction hypothesis they are independent, so every $c_j - c_{j-1} = 0$ for $j = 2, \dots, m$ (the denominators are positive, since no $d + 1$ consecutive knots of

Eq. 43.2.1 coincide). Hence all c_j are equal to some c , and $0 = s = c \sum_j B_j^d = c$ on $[a, b]$ by (c). So $c = 0$. ■

Idea. Local support is the whole point. Since only $d + 1$ B-splines are nonzero at any x , the matrix $\mathbf{B}$ has $d + 1$ nonzero entries per row and $\mathbf{B}^\top \mathbf{B}$ is banded of bandwidth d , and moving one coefficient γ_j changes the curve only on $[\kappa_j, \kappa_{j+d+1})$. Neither statement holds for the truncated power basis, where changing the coefficient of $(x - t_1)_+^d$ moves everything to the right of t_1 .

43.2.3 The coefficients live at the Greville abscissae

A spline written in the B-spline basis, $f = \sum_j \gamma_j B_j^d$, has coefficients that are not values of f , but are close to being so, and sit at identifiable places.

Proposition 43.2.2 — Greville abscissae

Let $\xi_j = (\kappa_{j+1} + \dots + \kappa_{j+d})/d$ for $j = 1, \dots, m$, the **Greville abscissae** of the sequence. For Eq. 43.2.1,

$$\sum_{j=1}^m \xi_j B_j^d(x) = x \quad \text{for all } x \in [a, b]. \quad (43.2.5)$$

Consequently $\sum_j (\alpha + \beta \xi_j) B_j^d(x) = \alpha + \beta x$: coefficients that are an affine function of the Greville abscissae produce an affine function of x .

Proof. Let $s(x) = \sum_j \xi_j B_j^d(x)$. Since $\xi_j - \xi_{j-1} = (\kappa_{j+d} - \kappa_j)/d$, Eq. 43.2.4 gives $s'(x) = \sum_{j=2}^m B_j^{d-1}(x) = 1$ on $[a, b]$, by the partition of unity of degree $d - 1$ established in the proof of Proposition 43.2.1(c). So $s(x) = x + c$ for a constant c , and s is continuous for $d \geq 2$ (for $d = 1$ the identity is immediate). At $x = a$ only B_1^d is nonzero by Proposition 43.2.1(a), so $B_1^d(a) = 1$ by the partition of unity, and $s(a) = \xi_1 = (\kappa_2 + \dots + \kappa_{d+1})/d = a$. Hence $c = 0$. The second statement follows by adding α times the partition of unity. ■

Remark (A uniform knot sequence). Eq. 43.2.1 is not the only choice. Penalized splines (Section 43.3) use instead a sequence that continues the equal spacing $\delta = (b - a)/(K + 1)$ of the interior knots d steps beyond each end: $a - d\delta, \dots, a - \delta, a, t_1, \dots, t_K, b, b + \delta, \dots, b + d\delta$. There are again $K + 2d + 2$ knots and $m = K + d + 1$ B-splines, and restricted to $[a, b]$ they span the same spline space, so every fitted curve is identical. The advantage is that the Greville abscissae are then equally spaced, $\xi_j = a + \{j - (d + 1)/2\}\delta$, consecutive ones δ apart, which is what makes the difference penalty of Section 43.3 behave exactly as advertised. That this sequence too has the partition of unity on $[a, b]$, gives a basis of

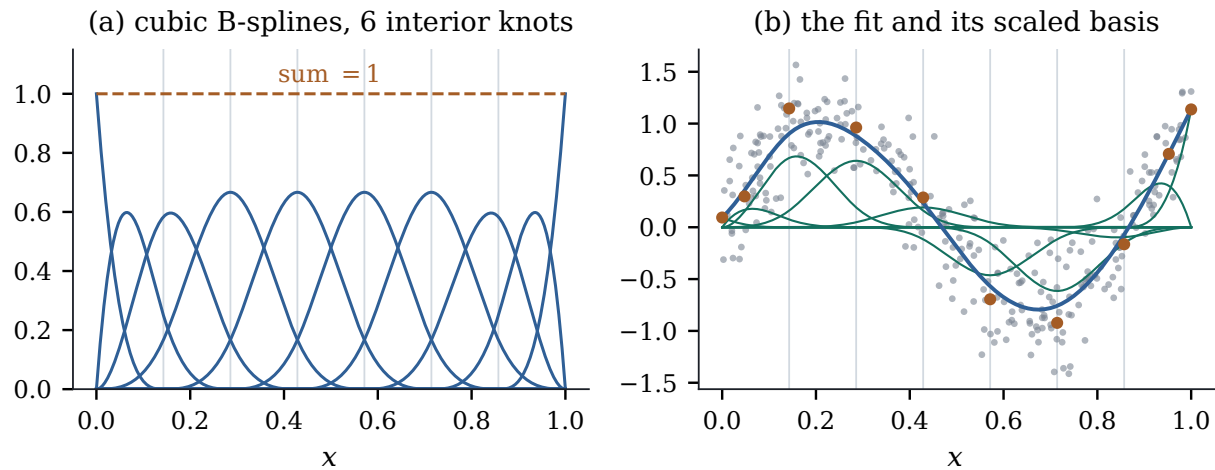


Figure 43.2.1: (a) The ten cubic B-splines on $[0, 1]$ with six equally spaced interior knots (vertical lines); the dashed line is their sum (Proposition 43.2.1(c)). (b) A least squares fit in this basis: thin curves $\gamma_j B_j^3$, thick curve their sum, dots the coefficients γ_j at the Greville abscissae of Proposition 43.2.2.

the spline space there, and satisfies Eq. 43.2.5 is quoted from de Boor (2001, chs. IX–X) — the proofs of (a), (c) and (f) above used the repeated endpoints — and confirmed numerically in Example (Two bases for the same space).

43.2.4 Conditioning

Example. (Two bases for the same space)

Take $n = 300$ design points drawn uniformly on $[0, 1]$, cubic splines ($d = 3$) and $K = 6$ equally spaced interior knots, so the spline space has dimension 10. Least squares in the truncated power basis and least squares in the B-spline basis give fitted values that agree to 1.5×10^{-13} , and the uniform sequence of the remark above agrees with the clamped one to 1.8×10^{-15} : three bases, one fit. The condition numbers Definition 1.8.1 of the two design matrices do not agree at all — here 5.3 and 2.1×10^4 . The table shows how they move with the number of knots.

K	$\kappa(\mathbf{B})$, B-spline	κ , truncated power
5	5.3	1.25×10^4
10	5.8	1.08×10^5
20	5.5	1.01×10^6
40	7.4	1.45×10^7

Figure 43.2.1 draws the basis and a fit in it. The B-spline condition number is essentially constant; the truncated power one grows by a decade for every doubling of the knot count, and squaring it to form a cross-product matrix costs twice as many digits again (Section 6.10). At $K = 40$ the truncated power fit has lost most of double precision, the B-spline fit a digit.

The whole recurrence is a dozen lines of numpy.

```

import numpy as np

def knot_vector(a, b, K, d):
    """d+1 copies of a, K equally spaced interior knots, d+1 copies of b."""
    interior = a + (b - a) * np.arange(1, K + 1) / (K + 1)
    return np.concatenate([np.full(d + 1, a), interior, np.full(d + 1, b)])

def bspline_basis(x, kn, d):
    """The B-splines of degree d on the knot vector kn, by the de Boor recurrence.
    """
    x = np.atleast_1d(np.asarray(x, float))
    B = np.array([(x >= kn[j]) & (x < kn[j + 1]) for j in range(len(kn) - 1)], float)
    B.T
    last = np.max(np.nonzero(kn[:-1] < kn[1:])[0]) # close the rightmost
    interval
    B[x == kn[-1], last] = 1.0
    for deg in range(1, d + 1):
        C = np.zeros((len(x), B.shape[1] - 1))
        for j in range(C.shape[1]):
            if kn[j + deg] > kn[j]: # a term with a zero
                denominator
                C[:, j] += (x - kn[j]) / (kn[j + deg] - kn[j]) * B[:, j]
                if kn[j + deg + 1] > kn[j + 1]: # contributes nothing
                    C[:, j] += (kn[j + deg + 1] - x) / (kn[j + deg + 1] - kn[j + 1]) * B
           [:, j + 1]
            B = C
    return B

D, K, A, Bnd = 3, 6, 0.0, 1.0 # cubic, six interior knots
    on [0, 1]
kn = knot_vector(A, Bnd, K, D)

```

[Run this code online](#)

```

def truncated_power(x, interior, d):
    """The basis 1, x, ..., x^d, (x - kappa_1)_+^d, ..., (x - kappa_K)_+^d."""
    x = np.asarray(x, float)
    powers = [x**j for j in range(d + 1)]
    return np.column_stack(powers + [np.maximum(x - k, 0.0) ** d for k in interior])

rng = np.random.default_rng(43043)
n = 300
x = np.sort(rng.uniform(0, 1, n))
y = np.sin(7 * x) + 0.4 * x**2 + 0.3 * rng.normal(size=n)

interior = kn[D + 1:D + 1 + K]
Bx, Tx = bspline_basis(x, kn, D), truncated_power(x, interior, D)
fit_b = Bx @ np.linalg.lstsq(Bx, y, rcond=None)[0]
fit_t = Tx @ np.linalg.lstsq(Tx, y, rcond=None)[0]
print("largest difference between the two fits:", np.max(np.abs(fit_b - fit_t)))
print("condition numbers:", np.linalg.cond(Bx), np.linalg.cond(Tx))

```

[Run this code online](#)

43.2.5 Choosing the knots, and why we shall stop choosing

With a B-spline basis, an unpenalized **regression spline** is ordinary least squares: $\hat{\mathbf{y}} = (\mathbf{B}^\top \mathbf{B})^{-1} \mathbf{B}^\top \mathbf{y}$, with $d+1+K$ parameters and the whole of Parts I–III available for its inference. Nothing is nonparametric about it. The difficulty is the one Proposition 42.4.1 describes: the fit depends on K and on where the knots are, the dependence is not smooth, and the selection machinery of Chapter 29 is then applied to a discrete search whose answer is unstable. Quantile-spaced knots are better than equally spaced ones when the design is uneven, but no placement rule removes the problem: the fit is a step function of K . Section 43.3 takes the opposite route — fix K at more knots than the data can support and control the fit with a penalty — so that a continuous dial takes the place of a discrete one.

43.2.6 Exercises

43.2.6.1 A. Check your understanding

Exercise A1.

For $d = 3$ and $K = 6$ interior knots, write out the clamped knot sequence Eq. 43.2.1 and check that it has $K + 2d + 2$ entries and yields $m = K + d + 1$ B-splines. Which B-splines are nonzero at $x = a$? At the midpoint of $[\kappa_3, \kappa_4]$?

43.2.6.2 B. Practice

Exercise B1.

Using the cell above, verify numerically for $d = 3$, $K = 6$ that $\sum_j B_j^3(x) = 1$, that B_j^3 vanishes outside $[\kappa_j, \kappa_{j+4}]$, that at most four B-splines are nonzero at any point, and that Eq. 43.2.5 holds. Then check Eq. 43.2.3 against a central difference.

Exercise B2.

Show that $\mathbf{B}^\top \mathbf{B}$ has $(\mathbf{B}^\top \mathbf{B})_{jk} = 0$ whenever $|j - k| > d$, and deduce that a Cholesky factorization (Theorem 10.2.1) of $\mathbf{B}^\top \mathbf{B}$ costs $O(nd^2)$ rather than $O(nm^2)$ operations. Why does the same statement fail for the truncated power basis?

Solution. $(\mathbf{B}^\top \mathbf{B})_{jk} = \sum_i B_j^d(x_i) B_k^d(x_i)$, and the supports $[\kappa_j, \kappa_{j+d+1})$ and $[\kappa_k, \kappa_{k+d+1})$ are disjoint when $|j - k| > d$, so every term is zero. The Cholesky factor of a band matrix inherits its bandwidth. Induct on j in Eq. 10.2.1: if $r_{ki} = 0$ whenever $i - k > d$ for every $k < j$, then for $i - j > d$ each product $r_{kj} r_{ki}$ in the sum vanishes — a nonzero r_{kj} needs $j - k \leq d$, whence $i - k > d$ — and $a_{ji} = 0$, so $r_{ji} = 0$. The factorization of Theorem 10.2.1 therefore visits $O(md^2)$ entries instead of $O(m^3)$, and forming the band of $\mathbf{B}^\top \mathbf{B}$ costs $O(nd^2)$. The truncated power functions have nested, not local, supports — the last is nonzero on $(t_K, b]$ and the first everywhere — so no entry of their cross-product matrix is structurally zero.

43.2.6.3 C. Going deeper

Exercise C1.

Suppose an interior knot is repeated ν times, $1 \leq \nu \leq d + 1$. Show from [Eq. 43.2.3](#), by induction on ν , that the B-splines are $C^{d-\nu}$ there, and that $\nu = d + 1$ allows a jump. How would you use this to fit a curve that is smooth everywhere except at one known breakpoint, where it is only continuous?

Exercise C2.

Prove that $\sum_j \psi_j(y) B_j^d(x) = (y - x)^d$ for all $x \in [a, b]$ and all y , where $\psi_j(y) = (y - \kappa_{j+1}) \cdots (y - \kappa_{j+d})$, for $d = 0$ and $d = 1$. (This is Marsden's identity; the general case is de Boor, 2001, ch. IX. Differentiating it $d - 1$ times in y and setting $y = x$ recovers [Eq. 43.2.5](#).)

Solution. For $d = 0$ the claim is $\sum_j B_j^0(x) = 1$, the empty product being one. For $d = 1$, $\psi_j(y) = y - \kappa_{j+1}$, and on $[\kappa_i, \kappa_{i+1})$ only B_{i-1}^1 and B_i^1 are nonzero, with values $(\kappa_{i+1} - x)/(\kappa_{i+1} - \kappa_i)$ and $(x - \kappa_i)/(\kappa_{i+1} - \kappa_i)$; their combination is

$$\frac{(y - \kappa_i)(\kappa_{i+1} - x) + (y - \kappa_{i+1})(x - \kappa_i)}{\kappa_{i+1} - \kappa_i} = y - x,$$

after expanding, since the terms in $\kappa_i \kappa_{i+1}$ cancel.